Validate Binary Search Tree (medium)

Problem Statement

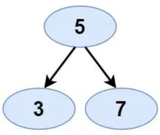
Determine if a given binary tree is a binary search tree (BST). In a BST, for each node:

- All nodes to its left have values less than the node's value.
- All nodes to its right have values greater than the node's value.

Example Generation

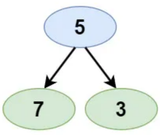
Example 1:

- **Input**: [5,3,7]
- **Expected Output**: true
- **Justification**: The left child of the root (3) is less than the root, and the right child of the root (7) is greater than the root. Hence, it's a BST.



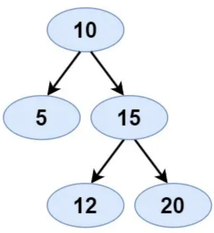
Example 2:

- **Input**: [5,7,3]
- **Expected Output**: false
- **Justification**: The left child of the root (7) is greater than the root, making it invalid.



Example 3:

- **Input**: [10,5,15,null,null,12,20]
- **Expected Output**: true
- **Justification**: Each subtree of the binary tree is a valid binary search tree. So, a whole binary tree is a valid binary search tree.



Try it yourself

Try solving this question here:

Java

```java
// class TreeNode {
//     int val;
//     TreeNode left;
//     TreeNode right;
//     TreeNode(int x) { val = x; }
// }

public class Solution {
    // Main method to check if a tree is a BST
    public boolean isValidBST(TreeNode root) {
        // ToDo: Write Your Code Here.
        return false;
    }
}
```

Test Case | Run Code Result | Stdout

[5,3,7]

Use Example Testcases | Run | Submit